GAT-CTDE: A Correctness-Aware Graph-Attention Architecture for Federated Multi-Agent Slice Admission Control

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Abstract

Fifth generation (5G) network slicing must admit heterogeneous service requests across many cooperating base stations under strict per-slice guarantees. This paper presents GAT-CTDE, a correctness-aware graph-attention architecture for federated multi-agent slice admission control. A shared Graph Attention Network (GAT) encoder reads the base-station cluster graph and feeds decentralised per-slice action heads. The policy trains centrally and executes locally, so each gNodeB (gNB) decides independently from a shared view of the cluster. The design closes three gaps identified in our prior systematic review, namely trustworthy evaluation, privacy-preserving coordination, and validated deployment. On the metric that reflects the training objective, GAT-CTDE decisively outperforms an independent-learner baseline. Scaled from 3 to 19 gNBs, it resists a training collapse that overwhelms a flattened single-agent baseline. It converges across 64 independent seeds to a stable 35.4% [25.5%, 45.8%] residual rate already present at 7 gNBs. Federated training adds no measurable reward cost, and differential privacy degrades decision quality only beyond a clear threshold. A correctness-aware metric pair and a reproduction-versus-replication discipline keep every reported result trustworthy. A live deployment on OpenAirInterface (OAI) hardware confirms that the offline-trained policy executes correctly end to end, establishing deployability on a real Open Radio Access Network stack. Contention-dependent behaviour is characterised in a disclosed offline stress regime calibrated harder than the live rig reaches.

Keywords: Network Slicing, Slice Admission Control, Multi-Agent Reinforcement Learning, Graph Attention Networks, Federated Learning, Open RAN

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1 Introduction

Network slicing lets one physical radio access network (RAN) serve several service classes at once. These classes include Ultra-Reliable Low-Latency Communication (URLLC), enhanced Mobile Broadband (eMBB), and massive Machine-Type Communication (mMTC). Each class carries its own per-slice Service Level Agreement (SLA). Our earlier work [7, 8] showed that a Deep Q-Network (DQN) admission controller can enforce these SLAs under a single gNodeB (gNB). We validated that controller on a live OpenAirInterface (OAI) testbed. A real deployment spans many cells. Neighbouring cells share backhaul, spectrum planning, and a cluster-wide load-balance objective, so admission at one gNB is never isolated. Coordinated admission is therefore a multi-agent problem. This raises a direct question. Does a shared, graph-structured view of the cluster let per-gNB agents decide better than a single agent on a flattened state, or than independent agents with no shared view? That question motivates this paper. Fig. 1 places the setting in context.

This paper's contribution is GAT-CTDE, a centralised-training, decentralised-execution multi-agent architecture. A shared Graph Attention Network (GAT) [17] encoder runs over the gNB topology and feeds decentralised per-slice Q-heads, delivered together with the evaluation discipline needed to trust what it reports. Three gaps from our systematic review of 67 core studies motivate this pairing, as Section 2 details. First, no reviewed study checks whether its own outcome metric can score a degenerate policy as well as a correct one. Second, none replicates a headline architectural finding on an independent seed sample. Third, none tests whether a live testbed can reach the contention regime its offline claims assume. We close all three gaps through a per-component realisation of the architecture, namely the GAT encoder, the Multi-Agent Reinforcement Learning (MARL) controller, the Federated Learning (FL) coordinator, and the Differential Privacy (DP) layer, each measured and isolated rather than folded into one end-to-end number. We apply a correctness-aware metric pair and a reproduction-versus-replication discipline throughout, and we add a live single-gNB deployment that tests, rather than assumes, whether the rig reaches genuine per-slice scarcity.

We then extend the architecture to federated training with differential privacy, and evaluate robustness under mid-episode gNB dropout, demand spikes, and agent churn. We scale the cluster to $N \in \{7, 19\}$ under ring and hex-grid adjacency, where GAT-CTDE resists a training collapse that becomes total for a flattened single-agent baseline. It converges to a residual 35.4% collapse rate across 64 independent seeds, rather than to either of two disagreeing small-sample estimates, and this rate is present from N=7 onward rather than only at large cluster size. On the metric that reflects the training objective, the architecture decisively beats an independent-learner ablation that isolates the same coordination contribution, although its edge over the single-agent baseline is small and not statistically decisive at this sample size, a limit we report honestly. Federated training costs nothing measurable in reward, differential

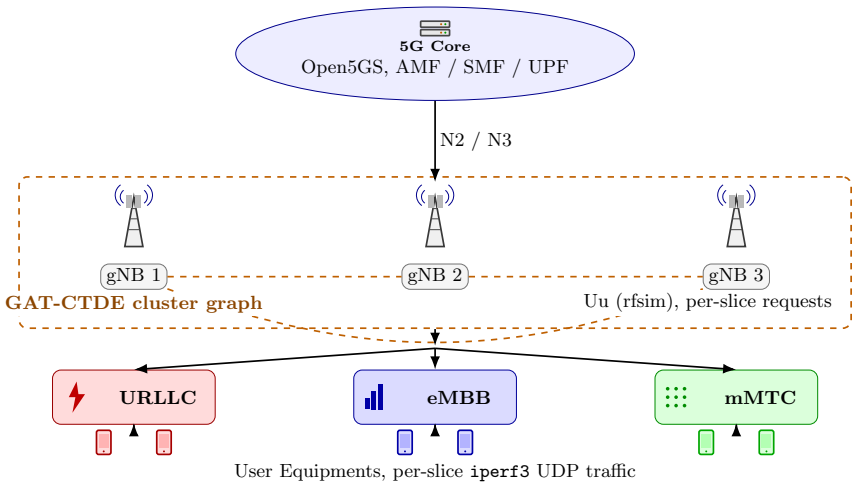


Fig. 1 System view. A cluster of gNBs, coordinated by the shared GAT-CTDE encoder over the cluster graph and backhauled to a 5G core, serves UEs across the URLLC, eMBB, and mMTC slices. Each gNB admits or rejects per-slice requests through a PRB-ratio ceiling.

privacy degrades decision precision only past a real threshold, and under disruption every arm's cost accelerates with severity.

These safeguards earn their place in practice rather than in principle. The correctness-aware pair diagnosed and fixed a training collapse in our first encoder across all 30 campaign seeds, as Section 4.1 describes, and the reproduction-versus-replication discipline corrected one architectural reading under disruption, as reported in Section 5.3. The live deployment confirms that the offline-trained decision logic executes correctly on real hardware, and it also showed that the rig cannot reach genuine per-slice scarcity at any tested load, so we make contention-dependent claims only in the disclosed offline stress regime.

2 Related Work

Our prior systematic review [9] screened 6,286 records down to 67 core studies from 2024 to 2026 on machine-learning-driven, user-centric distributed network slicing. The review organised them around three enabling pillars, namely topology-aware graph representation, distributed multi-agent control, and privacy-preserving federation. Each pillar is well developed on its own, but the three are rarely combined under realistic constraints, and never with a validated live deployment. No reviewed study checks whether its own evaluation metric can mask a degenerate policy, and none replicates a headline finding on an independent seed sample, which is the gap this paper fills.

Several recent works come close on a single axis of this problem. Shao et al. [14] motivate our GAT encoder choice by showing that graph attention strengthens inter-agent cooperation over plain DQN and Advantage Actor-Critic (A2C) baselines. Ahmadi et al. [2] study topology generalisation through graph attention, which complements but does not answer the fixed-topology evaluation-integrity question we ask

in Section 5.4. Sulaiman et al. [15] are the closest prior art to our joint slicing-and-admission-control problem. None of these three checks whether its reward-optimal policy could be mistaken for a degenerate one, and none replicates a headline result on an independent seed sample.

Robustness studies share our concern but stop short of it. Tashman and Cherkaoui [16] and Fatehi et al. [5] study robustness to a disruptive condition, as our disruption campaign in Section 5.3 does, yet neither checks whether the reported robustness metric is trustworthy and neither replicates a finding on an independent sample. That is the gap our independent-seed replication closes, and the need is concrete, since replication overturned our initial churn-immunity reading. Zhang et al. [19] are the closest periodic-synchronisation precedent, and Yasin et al. [18] are the closest precedent for federated learning with differential privacy. Their threat model differs from ours, since they secure the controller rather than characterise a control policy's own privacy-utility trade-off, which our σ sweep in Section 5.2 reports, and neither adds a correctness-aware check on whether federating or privatising changed decision quality. A further, closely related set [3, 6, 12, 13] combines two of the three pillars, yet like every work above, each evaluates on its own reported outcome metric, computed once, on one seed or run.

Rather than adding a new point in the GAT, MARL, and FL design space, this paper fills that gap with a demonstration on GAT-CTDE, extended to federation, privacy, and disruption. It shows that two checks are both necessary, namely testing a metric's own trustworthiness and replicating a finding on an independent sample. Skipping either changes what a reader may believe the resulting numbers mean.

3 System Model and Problem Formulation

Fig. 2 orients the reader before the detail. A GAT-CTDE controller runs across four offline campaigns in a disclosed stress environment. Two correctness safeguards check every result. The offline-trained logic is then verified on the live single-gNB testbed of Fig. 3.

We consider a cluster of N=3 gNBs, each serving the same three slices, URLLC, eMBB, and mMTC, under a shared per-gNB Physical Resource Block (PRB) budget B. An admission decision is a PRB-ceiling adjustment through the `slicing_control_m` minimum and maximum ratio, rather than a literal per-flow accept or reject, since the real OAI xApp control interface exposes only ratio ceilings. No real multi-gNB physical topology is available to us, so we take the least assumption-laden choice and treat the cluster as fully connected, as a modelling choice rather than a measured fact. At each control step, every gNB holds zero or more pending admission requests, each tagged with a slice identity, and an agent decides accept or reject for each request independently, conditioned on the current cluster state. The discrete action space matches our single-gNB work exactly at `action_dim`=2, so any coordination benefit found here is due to the architecture and not a richer action space.

The step reward follows our prior single-gNB formulation. It has a per-slice service term net of SLA-violation penalties, a congestion charge on admitted volume, and a load-imbalance penalty across gNBs. The step reward r_t sums over slices indexed

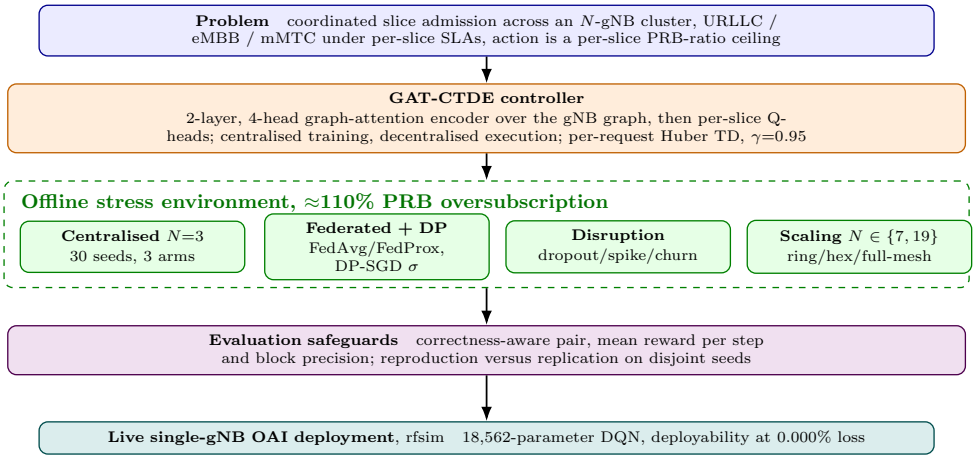


Fig. 2 Overview of the research. A GAT-CTDE controller is evaluated across four offline campaigns in a stress environment, every result checked by two safeguards, then validated on a live single-gNB testbed.

by k. It uses priority weight ω_k, per-accept reward R_k, admitted count n_k, violation penalty λ_k, and the indicator $\mathbf{1}[\cdot]$ equal to one exactly when slice k is in SLA violation that step,

$$r_t = \sum_k \big(\omega_k R_k\, n_k - \lambda_k\, \mathbf{1}[v_k]\big) \;-\; \kappa\, C_t \|a_t\|_1 \;-\; \beta\,(1 - F_t), \tag{1}$$

where $\|a_t\|_1$ is the total admitted this step, C_t the cluster-mean congestion level, κ the congestion coefficient, $F_t \in [0,1]$ the min-over-max PRB-utilisation fairness ratio, and β the load-balance weight. The violation flag v_k follows from a continuous SLA margin m_k. The correctness-aware metrics of Section 4.4 also build on this margin. We take it as the worse of the queue and loss channels,

$$m_k = \min\Big(1 - \tilde{q}_k,\; 1 - \tfrac{\ell_k}{\ell_k^{\max}}\Big), \qquad v_k \Leftrightarrow m_k \le 0, \tag{2}$$

with $\tilde{q}_k$ the slice backlog relative to its budget, and ℓ_k and $\ell_k^{\max}$ its loss proxy and loss budget. These weights are tuned in prior work so that, under congestion, the reward-optimal policy is differentiated shedding, which continues accepting URLLC, mostly accepts eMBB, and rejects mMTC. We confirm this by Q-value inspection on a working baseline policy in Section 4.1, rather than assume it. Every result uses an offline, closed-loop simulation, except the bounded live anchor of Section 5.5. The simulation is deliberately oversubscribed to roughly 110% of one gNB's PRB budget, and it reuses our existing multi-gNB configuration unmodified. We use it as a deliberately harder, disclosed stress environment for a contention regime the live rig cannot currently produce. At every native load tested, one and two User Equipments (UEs) per slice alike, all three slices serve essentially 100% of offered demand with zero loss, as Section 5.5 reports, so multi-gNB and disruption claims remain offline-only. The concrete rig behind every live measurement is the single-host, software-radio, Open Radio

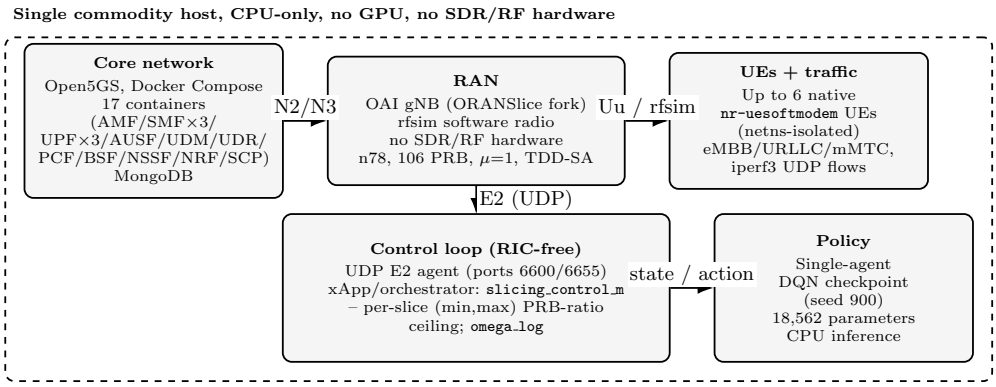


Fig. 3 Real testbed stack, a single-host, software-radio rfsim, RIC-free O-RAN deployment distinguished from SDR hardware-in-the-loop testbeds. Every block and label is a measured or configured value from this rig.

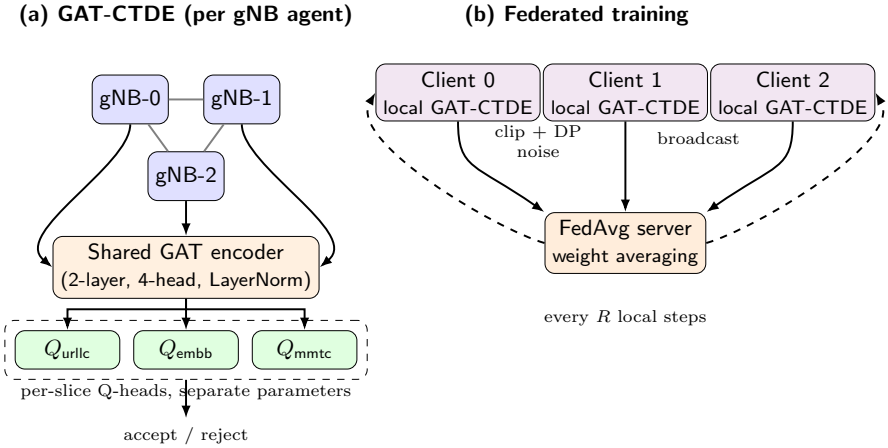


Fig. 4 GAT-CTDE architecture. a, A shared graph-attention encoder over the fully-connected 3-gNB topology produces one embedding per node, consumed at execution time by per-slice Q-heads using only that node's row. b, The federated variant, each gNB a client synchronised by periodic FedAvg with an optional DP-SGD clip-and-noise step.

Access Network (O-RAN) stack of Fig. 3, which runs without a near-real-time RAN Intelligent Controller (RIC) and which the live deployment of Section 5.5 exercises end to end.

4 GAT-CTDE Method

4.1 Shared Encoder, Decentralised Q-Heads

Fig. 4a shows the architecture. At every control step, each gNB forms its own per-slice feature row x_i. The features are PRB-used-ratio, congestion-level, and queue-length-norm. These rows stack into a joint node-feature matrix $X \in \mathbb{R}^{N\times d}$ over the N gNB nodes, with d features each. A two-layer, H=4-head GAT [17] produces one embedding per gNB. Each head h decomposes additive attention rather than scoring a concatenation. It uses a learned source vector a_s^h and destination vector a_d^h, applied to the shared linear projection W^h. Here ϕ is the leaky rectifier of slope 0.2, and $\mathcal{N}_i$ is the set of neighbours of node i under the adjacency mask,

$$\varepsilon_{ij}^h = \phi\big(a_s^{h\top} W^h x_i + a_d^{h\top} W^h x_j\big), \qquad \alpha_{ij}^h = \frac{\exp \varepsilon_{ij}^h}{\sum_{l\in\mathcal{N}_i} \exp \varepsilon_{il}^h}. \tag{3}$$

Here ε_{ij}^h is the unnormalised attention score from node i to neighbour j in head h, and α_{ij}^h is its normalisation over i's neighbours, with l ranging over $\mathcal{N}_i$. Hidden layers concatenate the H heads under an ELU nonlinearity, while the final layer averages them. A per-layer normalisation operator $\mathrm{LN}(\cdot)$ then rescales each node's embedding to zero mean and unit variance, and produces the final per-gNB embedding z_i,

$$z_i = \mathrm{LN}\Big(\tfrac{1}{H}\textstyle\sum_{h=1}^{H}\sum_{j\in\mathcal{N}_i} \alpha_{ij}^h\, W^h x_j\Big). \tag{4}$$

Centralisation comes from this encoder, whose parameters every agent's loss updates jointly, so it sees the full cluster state during training. Decentralised execution comes next, since at action-selection time each agent consumes only its own embedding row z_i to decide its own pending requests.

The `LayerNorm` and the per-slice heads both answer a real training collapse, rather than a theoretical preference. An early design fed each embedding, concatenated with a one-hot slice context, into a single shared Q-head multi-layer perceptron (MLP). All three slices then update the same shared parameters. URLLC's large priority weight produces large gradients that overwrite mMTC's much smaller true signal. This is a standard shared-parameter class imbalance, and it drove the evaluation policy to block zero requests. We replace the single shared head with one small MLP per slice, selected by the context one-hot. Each slice then owns its parameters, so one slice's gradients can no longer overwrite another's. Together with the `LayerNorm`, this reduced the collapse from 30 of 30 to 9 of 30 seeds. Correctly-targeted differentiated shedding rose from 0 to 21 of the 30 seeds. A seed here is one independent random initialisation of network weights and environment stream. We report every seed rather than a single best run.

Algorithm 1 states the training loop, where the encoder updates from every request in a sampled minibatch regardless of gNB or slice. This is centralised training, while action selection at rollout time consults only the acting agent's own embedding row, which is decentralised execution. Every pending request at gNB i, slice k is its own temporal-difference sample, and we never sum requests across a step before the loss.

Algorithm 1 GAT-CTDE: centralised training, decentralised execution

Require: encoder f_θ, per-slice Q-heads $\{Q_{\phi_k}\}$, targets $f_{\theta^-}, \{Q_{\phi_k^-}\}$, buffer $\mathcal{D}$, adjacency A, discount γ, sync period C

for each step t, observing joint node features X_t **do**
  $E_t \leftarrow f_\theta(X_t, A)$ {centralised, one row per gNB}
  for each pending request at gNB i, slice k, pick $a \leftarrow \arg\max_a Q_{\phi_k}(E_t[i])[a]$, ϵ-greedy {decentralised, own row only}
  step env, store $(X_t, X_{t+1}, \{(i,k,a)\}, r_t, \text{done})$ in $\mathcal{D}$
  if $|\mathcal{D}| \geq$ warmup **then**
    sample minibatch B; $E_B \leftarrow f_\theta(X_B, A)$, $E'_B \leftarrow f_{\theta^-}(X'_B, A)$
    per request, $y \leftarrow r + \gamma(1-\text{done}) \max_{a'} Q_{\phi_k^-}(E'_B[i])[a']$
    $\mathcal{L} \leftarrow$ mean per-request Huber loss, never summed per step; update $\theta, \{\phi_k\}$, clip grad-norm 1.0
  end if
  every C episodes, $\theta^- \leftarrow \theta$, $\phi_k^- \leftarrow \phi_k$
end for

For a request with immediate reward r, next-state embedding z'_i, and candidate next action a', the bootstrapped target y at discount γ and the minibatch loss are

$$y = r + \gamma\,(1-d)\max_{a'} Q_{\phi_k^-}(z'_i)[a'], \qquad \mathcal{L} = \frac{1}{|B|}\sum_{(i,k,a)\in B} \ell_\delta\Big(Q_{\phi_k}(z_i)[a] - y\Big), \tag{5}$$

where Q_{ϕ_k} is slice k's online Q-head, $Q_{\phi_k^-}$ its target copy, d the episode-done flag, and B the sampled minibatch of requests. We use the Huber loss ℓ_δ in place of squared error, which keeps the deeper encoder stable under long bootstraps.

4.2 Federated Training with Differential Privacy

Fig. 4b shows the federated variant. Each gNB becomes a federated-learning client. It holds a private copy of the same network and trains only on its own requests. Transitions are partitioned by owning agent, so no client sees another's raw admission data. Each client's GAT encoder still consumes the full joint node-feature snapshot. Every R local steps, the server averages all clients' online weights by FedAvg [11] and broadcasts the result back. We also implement FedProx [10], which adds a proximal term that pulls local weights toward the last broadcast global model. We sweep it at $\mu \in \{0.01, 0.1, 1.0\}$ under both homogeneous and heterogeneous per-gNB load. FedProx earns no measurable dividend at any strength, since per-round client drift is negligible relative to the loss scale.

Differential privacy gives a formal guarantee that one client's training data cannot be inferred from the shared model. We apply it as a standard differentially private stochastic gradient descent (DP-SGD) step [1]. Each client clips its gradient to a fixed norm C_{dp}. When privacy is enabled, the client then perturbs the gradient with Gaussian noise scaled by a noise multiplier σ, before its optimiser step. We report σ as the primary privacy parameter. Alongside it we report a formal (ε, δ) bound

Algorithm 2 Federated GAT-CTDE round with DP-SGD

Require: N clients with local nets $(\theta_i, \{\phi_{i,k}\})$, global snapshot $\bar{\theta}$, local steps R, clip norm C_{dp}, noise σ, FedProx weight μ

for each round **do**
  for R local steps, each client i independently **do**
    $\mathcal{L}_i \leftarrow$ per-request Huber TD loss of Alg. 1 on i's own requests $+ \frac{\mu}{2}\|\theta_i - \bar{\theta}\|^2$
    $g_i \leftarrow \text{clip}(\nabla\mathcal{L}_i, C_{\text{dp}})$, then $g_i \leftarrow g_i + \mathcal{N}(0, (\sigma C_{\text{dp}})^2 I)$ if $\sigma > 0$ {same clip isolates privacy cost}
    $(\theta_i, \phi_i) \leftarrow (\theta_i, \phi_i) - \eta\, g_i$
  end for
  $\bar{\theta}, \bar{\phi} \leftarrow \frac{1}{N}\sum_i (\theta_i, \phi_i)$ and broadcast to every client {FedAvg}
end for

via zero-concentrated differential privacy (zCDP) [4]. Here ε and δ are the standard privacy-loss and failure-probability parameters. Composing the Gaussian mechanism over m per-client gradient releases gives the zCDP parameter ρ and the bound

$$\rho = \frac{m}{2\sigma^2}, \qquad \varepsilon(\delta) = \rho + 2\sqrt{\rho \ln(1/\delta)}. \tag{6}$$

We apply the same clip whether or not noise is added, so a σ=0 run and a σ>0 run differ only in whether noise is injected, and the privacy-cost comparison in Section 5.2 depends on this isolation. Algorithm 2 states one federated round, in which the proximal weight μ is the FedProx strength swept above.

4.3 Experimental Setup

Both campaigns use 300 training and 50 evaluation episodes per seed. The centralised campaign uses a fixed 30-seed list, seeds 900 to 929, across three arms. The first arm is GAT-CTDE. The second is a single-agent DQN baseline run against the flattened joint state. The third is an independent-DQN ablation. An ablation is a deliberately reduced variant with one component removed, so its effect can be isolated. Here it is one per-gNB DQN with no parameter sharing and no topology information, which isolates the contribution of the shared encoder and centralised training. All three arms use matched hyperparameters, with discount factor $\gamma = 0.95$ and per-episode epsilon decay. The federated campaign uses the first 10 of these seeds and a five-level noise sweep $\sigma \in \{0, 0.5, 1, 2, 4\}$. Its σ=0 level is therefore directly comparable to the same 10 centralised seeds. All confidence intervals (CIs) are 95% bootstrap percentile intervals over 10,000 resamples. Paired comparisons use the Wilcoxon signed-rank test.

4.4 Evaluation Safeguards

Two safeguards make the results below trustworthy. Table 1 summarises the four pipeline issues they caught. A standard pipeline would have surfaced each one as a clean result. The first safeguard is a correctness-aware metric pair, computed from data every run already logs. Mean reward per step is the actual training objective.

Table 1 Four evaluation-pipeline failures this paper's safeguards caught, none of which a standard pipeline would have surfaced.

Failure	Missed by	Caught by	Outcome
Training collapse (§4.1)	Compliance blind to no-op	Correctness-aware pair	0/30→21/30 differentiate
Eval-log contamination	Append-mode log stacking	Episode-index audit	Logs cleared, no retrain
Reproducibility gap (§5.3)	Single-sample overclaim	Repro-vs-replication	Churn claim retracted
Scheduler-floor bug (§5.5)	Can't see cross-layer floor	Direct state instrumentation	>100k readings clear

Block precision is the fraction of blocked requests that correctly target mMTC, the only slice the reward calibration ever makes reject-optimal. Writing b_{mMTC} and b for the mMTC-targeted and total blocks over an evaluation,

$$\pi = \frac{b_{\mathrm{mMTC}}}{b}, \quad b > 0, \qquad \text{undefined for } b = 0, \tag{7}$$

a seed that never blocks is therefore excluded, rather than scored as a success.

The pair is needed because `sla_compliance_all_slices` scores worse when the policy becomes more correct. Blocking mMTC under congestion never recovers mMTC's own SLA margin in this stress regime. We re-checked every paired comparison in this paper against both metrics. The correctness-aware pair and compliance disagree on direction or significance in 44 of 49 comparisons, or 89.8%. A disjoint supplementary seed range confirmed this at 21 of 24. Every number in this paper that matters is therefore reported through the correctness-aware pair.

The second safeguard is a reproduction-versus-replication discipline, in which we recompute every headline comparison on an independent, disjoint seed sample drawn after the first and report both. Reproduction re-runs the same seeds, whereas replication draws fresh ones, so only replication can catch a result that is an artifact of one seed sample. Applied to the disruption campaign in Section 5.3, it corrected an early reading that independent DQN was immune to agent churn, which the committed sample alone had made look confirmed. The remaining two issues were handled directly at their source. A logging bug had stacked three runs' evaluation episodes in each seed's log, so we cleared the logs and re-evaluated against the already-trained checkpoints with no retraining, while a live scheduler-floor misconfiguration is resolved in Section 5.5.

5 Results

5.1 Centralised

The N=3 fully-connected topology is deliberately the setting where a shared graph-attention encoder has the least topology to exploit. This makes it a clean substrate for

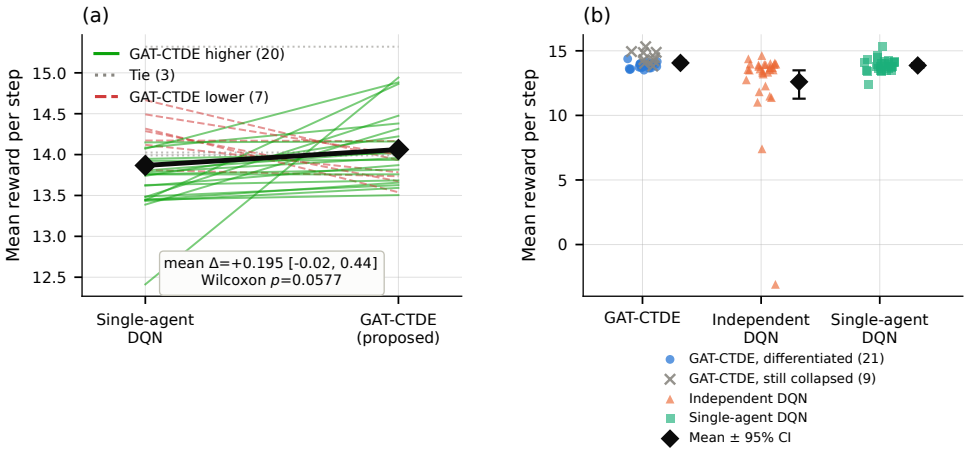


Fig. 5 M2 centralised campaign, per-seed evidence. a, Paired per-seed mean reward per step versus single-agent DQN, coloured by win, tie, or loss, the source of the borderline $p = 0.0577$. b, Per-seed reward distribution for all three arms.

Table 2 M2 campaign results, 30 seeds/arm, 95% bootstrap CIs.

Arm	Mean reward/step	Block precision	Compliance*
GAT-CTDE	14.062 [13.91, 14.23]	0.952	0.151 [0.083, 0.225]
Indep. DQN	12.601 [11.30, 13.49]	0.901	0.049 [0.013, 0.093]
Single-agent	13.867 [13.69, 14.05]	0.987	0.115 [0.061, 0.176]

* Low compliance is the metric artifact Section 4.4 explains, not a performance failure.

the evaluation-integrity findings, free of any topology effect. The encoder's coordination advantage is tested properly only once the cluster scales to N=19 under sparser adjacency in Section 5.4.

Table 2 and Fig. 5 report the final 30-seed campaign. On mean reward per step, the metric that reflects the training objective, GAT-CTDE beats the independent-DQN ablation decisively. The margin is +1.461, with 95% CI [0.550, 2.782] and $p < 0.0001$. This isolates a real contribution from the shared encoder and centralised training. Its edge over single-agent DQN is small and not statistically decisive at this sample size, at +0.195 with 95% CI [−0.017, 0.442] and $p = 0.0577$. Block precision is high at 0.952, so when GAT-CTDE blocks a request it is almost always the reward-correct slice. The compliance metric shows no clear direction, at +0.036 with $p = 0.4549$, an artifact that Section 4.4 explains.

5.2 Federated Training and DP

Comparing the centralised campaign against the federated variant with no DP noise, on the same 10 seeds, mean reward per step moves by +0.133, with 95% CI [−0.138,

Table 3 Training hyperparameters, all fixed across seeds.

Hyperparameter	Value
Learning rate (GAT-CTDE)	1×10^{-4}
Learning rate (DQN baselines)	1×10^{-3}
Discount γ	0.95
Epsilon: start $\rightarrow$ end (decay/episode)	1.0 $\rightarrow$ 0.05 (0.985)
Target-network sync	every 10 episodes
Optimiser	Adam
Training / evaluation episodes	300 / 50
GAT encoder, GAT-CTDE only	
Hidden / output dim	16 / 16
Attention heads	4
Encoder layers	2
Q-head hidden dim	32
Federated + DP, federated GAT-CTDE only	
Local steps per round	50
Aggregator	FedAvg (μ=0)
DP gradient clip norm	1.0
DP noise multiplier sweep σ	$\{0, 0.5, 1, 2, 4\}$

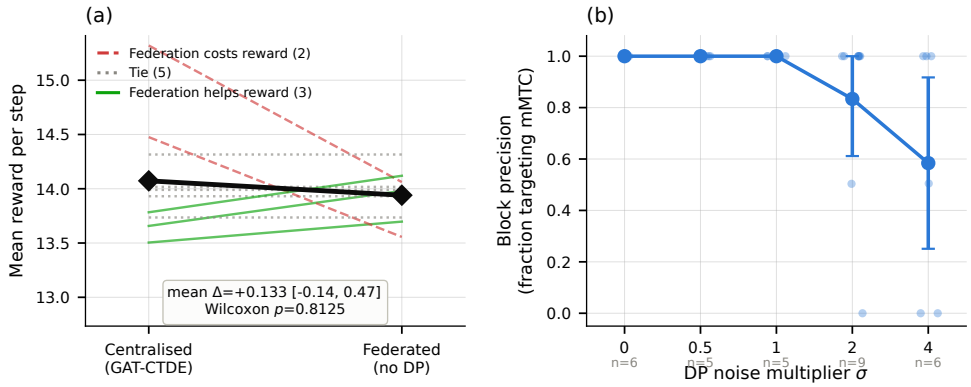


Fig. 6 M3 federated and DP sweep. a, Paired per-seed federation cost on mean reward per step, no measurable difference at $p = 0.8125$. b, Block precision versus DP noise multiplier σ, perfect through σ=1.0 then dropping sharply.

0.470] and $p = 0.8125$. Federating therefore carries no measurable cost, and both arms hold perfect block precision on seeds that block at all. Sweeping the DP noise multiplier in Fig. 6b, block precision stays perfect through σ=1.0, then drops sharply to 0.834 at σ=2.0 and 0.584 at σ=4.0, a real threshold rather than a smooth degradation. Mean reward per step stays within a narrow 12.45–14.19 band, since DP noise erodes which slice is blocked rather than whether the policy differentiates at all. Read with the FedProx null result, this answers a deployability question directly. Periodic, infrequent synchronisation suffices for this architecture, so a disaggregated RIC needs no per-step

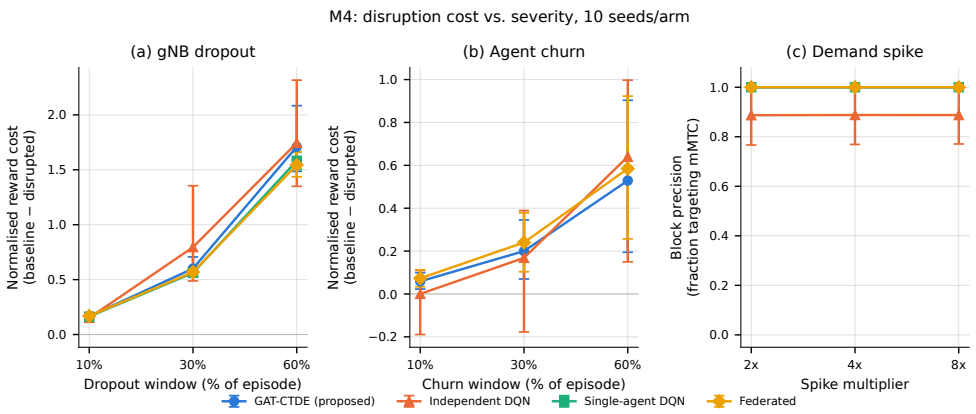


Fig. 7 M4 disruption campaign, 10 seeds per arm. a, b, Normalised reward cost versus severity for dropout and churn, accelerating with severity. c, Block precision versus demand-spike multiplier, flat and near-ceiling.

cross-gNB coordination to match centralised reward, and differential privacy can be added at a bounded, known precision cost.

5.3 Disruption

We evaluate every arm's frozen checkpoint against three mid-episode disruptions, with no retraining. We inject each one at evaluation time. 'gNB dropout' forces a gNB's requests to reject and zeroes its row in the joint feature matrix for a window. 'Demand spike' is a transient arrival-rate multiplier external to the policy. 'Agent churn' substitutes a gNB's decisions with a fresh, never-trained copy of its own policy class for a window, and leaves other agents untouched. Churn applies to every genuinely multi-agent arm, but not to single-agent DQN. For dropout and churn, severity is the window duration, at 10%, 30%, or 60% of the episode. For spike, severity is the arrival-rate multiplier, at 2×, 4×, or 8×. Raw reward inflates mechanically with request volume. We therefore report the volume-normalised mean reward $\bar{r}$ over an episode of T steps, dividing each step reward r_t by its own pending-request count n_t before averaging,

$$\bar{r} \;=\; \frac{1}{T}\sum_{t=1}^{T}\frac{r_t}{\max(n_t,\,1)}. \tag{8}$$

Every dropout and churn cost accelerates with severity rather than growing proportionally, with Wilcoxon $p \le 0.03$ and 9 or 10 of 10 seeds hurt in every case. On the committed sample, GAT-CTDE's dropout cost rises 0.165, 0.601, 1.710 across the three severities, and its churn cost rises 0.059, 0.200, 0.529. Every other arm shows the same accelerating shape. Demand spike does not corrupt decision quality on the volume-invariant block-precision metric in Fig. 7c. Its apparent per-step reward gain is a denominator artifact, not better decisions.

Independent DQN's churn cost needs both seed samples to state honestly. The committed sample is small and not conventionally significant, and an earlier pass read this as independent DQN being immune to churn, since it never attends to another agent's features. A disjoint independent-seed replication is significant at every severity and larger than the coordinated arms', with its cost rising 0.097, 0.657, 1.223 against the committed 0.000, 0.168, 0.640. The corrected reading is that churn always hurts, and that treating a borderline n=10 result as a confirmed property was the error. This is the one exception among four cross-section headline comparisons checked against an independent sample. The other three replicate with overlapping intervals, namely GAT-CTDE over independent DQN on reward at +1.461 then +0.767 with $p < 0.0001$ in both, GAT-CTDE over single-agent at +0.195 then +0.445, and a federation cost negligible in both.

5.4 Cluster-Size Scaling

We extend the cluster from N=3 to $N \in \{7, 19\}$ under three adjacency structures, fully-connected, ring, and hex-grid. We scale synthetic arrivals with N to hold per-gNB offered load roughly constant. GAT-CTDE is the only arm whose Q-head consumes this adjacency. Independent DQN and single-agent DQN therefore serve as a topology-invariance check. We normalise reward per gNB as in Eq. 8. We report block precision alongside, since compliance is OR-aggregated across gNBs and trends to zero as N grows, regardless of policy quality. The primary sample is 12 seeds at N=19 across all topologies and arms, with independent replication seeds for generalisation.

At N=19, the paired per-gNB reward margin is not significant for any topology, which is mechanically expected, since single-agent DQN's always-accept collapse collects the service term on every request while GAT-CTDE pays a real cost each time it correctly rejects mMTC. The advantage therefore lives in collapse resistance and block precision.

Single-agent DQN collapses totally at N=19, in 15 of 15 cells combined across both samples and every topology, confirmed at the individual-decision level by a direct Q-value probe. GAT-CTDE collapses only partially, while independent DQN never fully collapses but sits in a narrow, mediocre 0.29–0.49 block-precision band, identical across topologies, as expected for an arm that never reads adjacency.

GAT-CTDE's collapse rate converges across 64 independently-drawn seeds, pooling 68 of 192 cells, to

$$\hat{p}_{\text{collapse}} = 35.4\%, \qquad 95\%\ \text{CI}\ [25.5\%,\ 45.8\%], \tag{9}$$

a single seed-level-bootstrapped estimate, rather than either of two smaller disagreeing samples. A properly powered 12-seed resample at N=7 under the identical stress environment, with a disjoint 12-seed replication, puts GAT-CTDE at 41.7% and 58.3% collapse across the two samples. This is statistically indistinguishable from Eq. 9, so its collapse-proneness is present from N=7 onward rather than only at large cluster size. Over the same two samples, single-agent DQN collapses at 91.7% and 75.0% against its earlier three-seed pilot's 0%, the clearest single result in this extension,

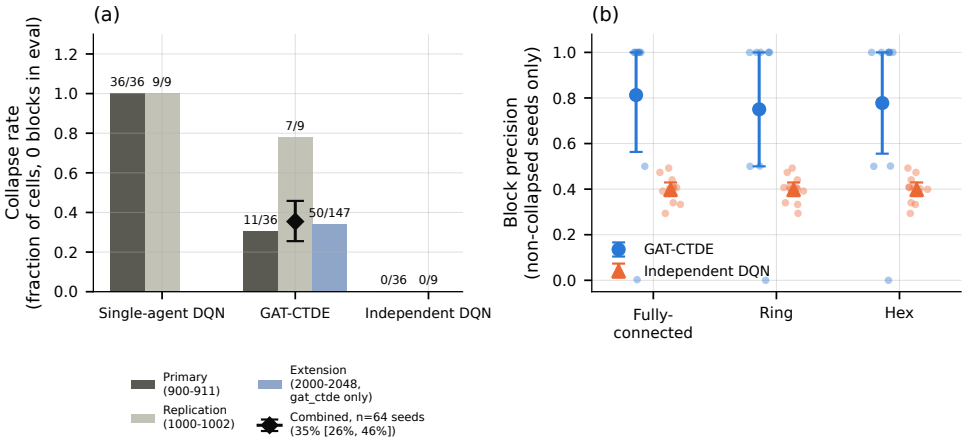


Fig. 8 N=19 cluster, per-seed evidence. a, Collapse rate pooled across topologies, single-agent DQN total and GAT-CTDE converging to 35.4% across 64 seeds. b, Block precision by topology for non-collapsed seeds.

while independent DQN collapses at 0.0% and 16.7%, which confirms it collapses rarely rather than never.

5.5 Live Deployment

This section reports what the real OAI testbed of Fig. 3 can and cannot establish about the offline-trained policy. Section 3 introduced that testbed. We use a single-agent-DQN checkpoint, seed 900, against a real gNB, real UEs, and real per-slice User Datagram Protocol (UDP) traffic. We test at native load of one UE per slice and at double native load of two UEs per slice.

A live capacity probe at a deliberately wide-open ceiling found that all three slices serve essentially 100% of offered demand with 0.000% loss at both loads. The frozen checkpoint's decision logic executes correctly end to end, with real traffic through the real E2 loop, real accept and reject decisions, and its characteristic low-priority-only blocking of mMTC reproducing at both loads, which establishes deployability. The rig never reaches scarcity, so the pattern is equally consistent with the policy's trained priority bias, and we therefore claim admission-control behaviour under contention only in the disclosed offline stress regime, not live. We also resolved two integrity issues during these checks. An early scheduler-configuration issue converted `saclb_live.yaml`'s per-slice PRB-ratio ranges to raw grants below the gNB scheduler's hard 5-PRB minimum, which we fixed at its root, after which direct instrumentation confirmed zero below-floor grants across more than 100,000 readings. A separate unit mismatch in the live SLA-margin diagnostic is fixed by converting backlog to a queueing-delay estimate via Little's Law, against each slice's own latency budget.

6 Conclusion

This paper presented GAT-CTDE, a correctness-aware graph-attention architecture for federated multi-agent slice admission control. A shared GAT encoder reads the cluster graph and feeds decentralised per-slice action heads. The design closes three gaps from our prior systematic review, namely trustworthy evaluation, privacy-preserving coordination, and validated deployment.

The architecture delivers on each of these fronts. On the metric that reflects the training objective, GAT-CTDE beats an independent-learner ablation decisively, although its edge over the single-agent baseline is smaller, which we report honestly. Federation adds no measurable reward cost, differential privacy stays accurate until a clear threshold, and disruption cost accelerates with severity for every arm. Scaled to 19 gNBs, GAT-CTDE resists a collapse that overwhelms the single-agent baseline, and it converges across 64 seeds to a stable 35.4% rate, with 95% CI [25.5%, 45.8%], already present at 7 gNBs. A live single-gNB deployment confirms that the offline-trained logic executes correctly on real hardware, which establishes deployability.

Two safeguards keep every one of these results trustworthy. A correctness-aware metric pair ties evaluation to the training objective, while a reproduction-versus-replication discipline tests each headline finding on an independent seed sample. Together they caught issues a standard pipeline would have reported as clean, and every headline number here passes through both. The broader lesson is practical, since evaluation in these settings needs metrics tied to the training objective, architectural claims need an independent seed sample before they are trusted, and a live testbed's reach must be checked rather than assumed.

A few limits bound the scope of these claims. The multi-gNB, federated, and disruption results are offline, the stress environment is deliberately harder than any load this rig reaches, the graph models only the gNB cluster with no compute-placement dimension, and the federated setting is a single-operator privacy scenario.

Future work targets the remaining gaps from our review. The first goal is a full-scale simulation with real contention, so admission-control behaviour under genuine scarcity can be measured directly rather than in the disclosed stress regime. The second is Multi-Access Edge Computing (MEC) scaling, which adds compute-placement nodes to the cluster graph and extends the controller to joint admission and offloading. We also plan a live multi-gNB deployment, broader privacy settings across operators, and direct Quality of Experience (QoE) inference in the reward. Together these steps move the architecture toward a scalable, privacy-aware, and user-centric controller for 5G and Beyond-5G networks.

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Declarations

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- **Competing interests:** The authors declare no competing interests.
- **Ethics approval and consent to participate:** Not applicable, since this study involves no human participants, animal subjects, or identifiable patient data. All reported results are either offline simulation campaigns or measurements from a laboratory radio testbed.
- **Consent for publication:** Not applicable, since this manuscript contains no personal data or identifying information about any individual.
- **Data availability:** The experiment logs, checkpoints, and campaign result files underlying this paper's findings are available from the corresponding author on reasonable request.
- **Materials availability:** Not applicable, since this study involves no physical or biological materials.
- **Code availability:** The framework and experiment scripts used to produce this paper's results, including the reproducibility manifest and analysis scripts used to independently re-derive every table and figure in this paper, are available from the corresponding author on reasonable request.
- **Author contribution:** M. P. Kunasegran: conceptualisation, methodology, software, investigation, writing (original draft). W. L. Pang: supervision, writing (review and editing). S. K. Phang: supervision, writing (review and editing).

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